Tissue Damage Analysis

8-hour simulation | STII model (Linder-Ganz & Gefen 2007)

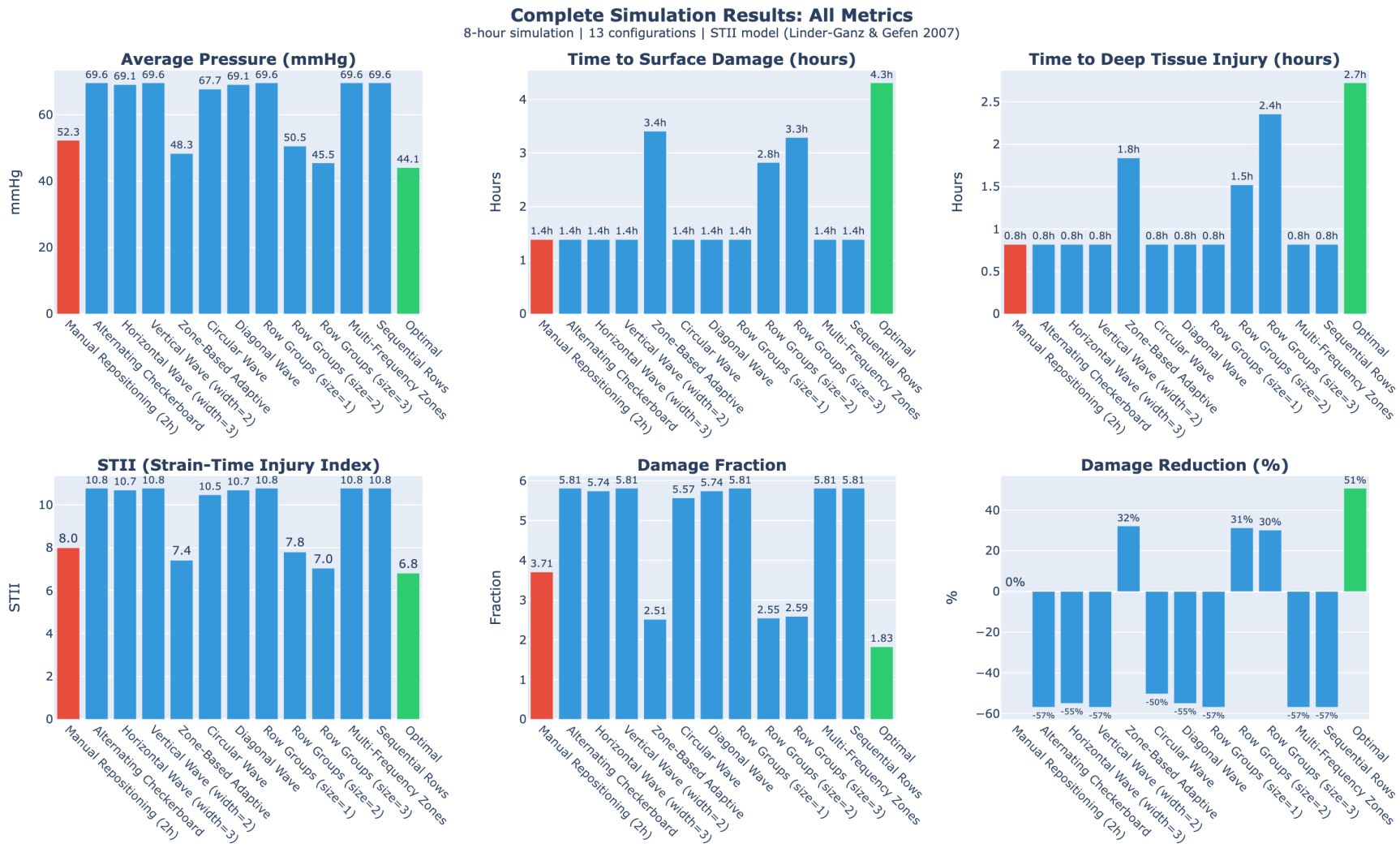

Table 1: Raw Simulation Data

Configuration	Pressure	Surface (h)	DTI (h)	STII	Damage
Manual Repositioning (2h)	52.26	1.4	0.8	8.00	3.71
Alternating Checkerboard	69.58	1.4	0.8	10.77	5.81
Horizontal Wave (width=3)	69.05	1.4	0.8	10.69	5.74
Vertical Wave (width=2)	69.58	1.4	0.8	10.77	5.81
Zone-Based Adaptive	48.29	3.4	1.8	7.42	2.51
Circular Wave	67.67	1.4	0.8	10.47	5.57
Diagonal Wave	69.07	1.4	0.8	10.69	5.74
Row Groups (size=1)	69.58	1.4	0.8	10.77	5.81
Row Groups (size=2)	50.52	2.8	1.5	7.80	2.55
Row Groups (size=3)	45.47	3.3	2.4	7.04	2.59
Multi-Frequency Zones	69.58	1.4	0.8	10.77	5.81
Sequential Rows	69.58	1.4	0.8	10.77	5.81
Optimal	44.08	4.3	2.7	6.81	1.83

Table 2: Percent Change from Baseline

Configuration	Pressure %	DTI %	STII %	Damage Red %
Manual Repositioning (2h)	Baseline	Baseline	Baseline	0.0
Alternating Checkerboard	+33.1	+0.0	+34.7	-56.8
Horizontal Wave (width=3)	+32.1	+0.0	+33.6	-55.0
Vertical Wave (width=2)	+33.1	+0.0	+34.7	-56.8
Zone-Based Adaptive	-7.6	+124.5	-7.3	32.2
Circular Wave	+29.5	+0.0	+30.9	-50.3
Diagonal Wave	+32.2	+0.0	+33.6	-55.0
Row Groups (size=1)	+33.1	+0.0	+34.7	-56.8
Row Groups (size=2)	-3.3	+85.7	-2.4	31.3
Row Groups (size=3)	-13.0	+187.8	-11.9	30.2
Multi-Frequency Zones	+33.1	+0.0	+34.7	-56.8

Sequential Rows	+33.1	+0.0	+34.7	-56.8
Optimal	-15.7	+232.7	-14.8	50.7

Best Configuration: Optimal

Metric	Value
Pressure Reduction	-15.7%
DTI Improvement	+232.7%
STII Reduction	-14.8%
Damage Reduction	50.7%
DTI Efficiency	333%